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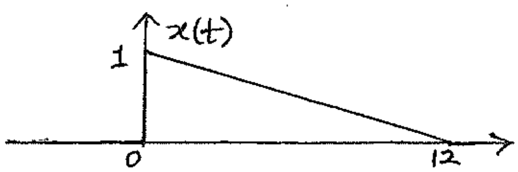
Third Semester B.Tech. (ECE/ECE-IoT)
End Semester Examination-2024

Course Code: ECECC302/EIECC302 Course Title: Signals and Systems

Time: 3:00 Hrs Max. Marks: 50

Note: Attempt all questions. Missing data/information (if any) may be suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
	Attempt any 02 parts of the following		
Q1(a)	Let $x(t)$ be the input and $y(t)$ be the output of a system. Determine whether it is (i) linear, (ii) time-invariant, (iii) causal, and (iv) stable. $y(t) = \begin{cases} 0, & x(t) < 0 \\ x(t), & x(t) \geq 0 \end{cases}$	5	CO1
Q1(b)	Determine whether the following signal is periodic, if yes, find its fundamental period. $x(n) = (\sqrt{j})^n$.	5	CO1
Q1(c)	For $x(t)$ indicated in Fig. 1, sketch the following. (i) $x(1-t)[u(t+1) - u(t-2)]$ (i) $x(1-t)[u(t+1) - u(2-3t)]$ Fig. 1 	5	CO1
	Attempt any 02 parts of the following		
Q2(a)	Determine if the following statement is true in general. Provide proof if you think it is true and counterexample if you think it is false. If $y(t) = x(t) * h(t)$, then $y(2t) = 2x(2t) * h(2t)$.	5	CO2
Q2(b)	(i) Using convolution, determine and sketch the response $y_1(t)$ of a linear, time-invariant system with impulse response $h(t) = e^{-t/2}u(t)$ to the step input $x_1(t) = u(t)$. (ii) Another input $x_2(t)$ can be expressed in terms of $x_1(t)$ as $$x_2(t) = 2[x_1(t) - x_1(t-3)]$$ By taking advantage of the linearity and time-invariance properties, determine how $y_2(t)$ can be expressed in terms of $y_1(t)$.	5	CO2
Q2(c)	Consider a discrete-time LTI system with impulse response $$h(n) = \sum_{k=-\infty}^{\infty} \delta(n - kN)$$ This system is not invertible. Find two inputs that produce the same output.	5	CO2

Q. No.	Question	Marks	CO
	Attempt any 02 parts of the following		
Q3(a)	Let $x(n)$ be a sequence with DTFT given by $X(e^{j\omega})$. Using the property of the DTFT determine the DTFT of the following sequence $g(n) = j^n x(n)$.	5	CO3
Q3(b)	Consider the signals $x(t) = \frac{\sin 4\pi t}{\pi t}$ and $y(t) = \cos(2\pi t)$. Find the Fourier transform of $g(t) = x(t)y(t)$. Sketch and clearly label the $G(\omega)$.	5	CO3
Q3(c)	Let$x(t)$ be a periodic signal with fundamental period T and Fourier series coefficients X_k. Derive the Fourier series coefficients of (i) $x(t) = x(T-t)$ and (ii) $Even\{x(t-T/2)\}$ in terms of X_k.	5	CO3
	Attempt any 02 parts of the following		
Q4(a)	An absolutely integrable signal $x(t)$ is known to have a pole at $s = 2$. Answer the following questions with justification: (i) Could $x(t)$ be of finite duration? (ii) Could $x(t)$ be left sided? (iii) Could $x(t)$ be right sided? (iv) Could $x(t)$ be two sided?	5	CO4
Q4(b)	A system impulse response $h(t)$ has a unilateral Laplace transform, $H(s) = \frac{s(s-4)}{(s+3)(s-2)}$ What is the region of convergence. Find the impulse response $h(t)$. Could the CTFT of $h(t)$ be found directly from $H(s)$? If not, why not?	5	CO4
Q4(c)	Consider a signal $y(t)$ which is related to two signals $x_1(t)$ and $x_2(t)$ by $y(t) = x_1(t-2) * x_2(-t+3)$. where $x_1(t) = e^{-2t}u(t)$ and $x_2(t) = e^{-3t}u(t)$ Use properties of the Laplace transform to determine the Laplace transform $Y(s)$ of $y(t)$.	5	CO4
	Attempt any 02 parts of the following		
Q5(a)	Consider a system with transfer function $H(z)$ (with impulse response $h(n)$ real & even). It is given that $H(z)$ has zero at $z = 2 + 2j$. What other minimum number of zeros must $H(z)$ have? Sketch the zero-diagram.	5	CO5
Q5(b)	A causal signal $x(n)$ has the z-transform $X(z) = \sin(z^{-1})(1 + 3z^{-2} + 2z^{-4})$. Find $x(11)$.	5	CO5
Q5(c)	Let $x(n) = (-1)^n u(n) + a^n u(-n - n_0)$. Determine the constraints on the complex number a and the integer n_0, given that the ROC of $X(z)$ is the region $1 < \lvert z\rvert < 2$.	5	CO5